Ilja Safronovs

R&D Engineer — 3D reconstruction & computer vision · Giraffe360 · Riga, Latvia

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I work on 3D reconstruction and computer vision: Gaussian splatting pipelines, benchmarking systems that measure every change, and continuous research that turns the best new papers into production code. Financial Engineering student at RTU.

EXPERIENCE

Giraffe360 — R&D Engineer — Riga · 2025 – present

- Solved the long-standing window, outdoor-view and mirror quality problem in 3DGS video. I found the root cause (wrong depth behind windows and walls), handled depth differently inside and outside the room, and added a training-time correction. 700+ controlled experiments back it. It runs in production and was publicly showcased by Giraffe360: instagram.com/p/DZsNRB8o772.
- Looped drone video, end to end: proposed the feature, agreed its look with the designers, developed the camera-path algorithm and shipped it to production. A customer-requested feature, launched by Giraffe360 in Oct 2026: instagram.com/giraffe360/reel/Dd9ODBVIHhE.
- Research to production: developed “influence accumulation”, which measures each Gaussian’s contribution to the image and relocates the least useful ones during training (in production). Implemented Analytic-Splatting (exact per-pixel Gaussian integration) in gsplat at CUDA level, which reduces aliasing; it is used in a downsampling algorithm.
- Benchmarking and evaluation: validation tools and ground-truth datasets, to measure quality against known truth; an experiment runner for agent-driven research loops on local GPUs; ClearML GPU runs.
- Led a small team that built exterior (outdoor) splats and orbital outdoor videos, now in production.
- Mentored a junior engineer who built a new depth algorithm with me (mostly his work): minima matching plus OMNI-DC to densify LiDAR depth. Now the best method; the team uses it.
- Led the pipeline restructuring and created a monorepo for all 3DGS extensions and client apps: ~3000-line files split, duplication removed, defects fixed. Enabled the half-floor camera fix.
- Embeddable Rust/wgpu splat viewer (host control, PLY comparison); multi-view comparison tool.
- Working prototypes: mirror planes with correct reflections; a fly-in “dollhouse” video intro.
- Mesh generation from splats, combined with Gaussian splats; point-count reduction for viewers.
- Diffusion-based artifact fixing (Difix, NVIDIA Fixer) made runnable on shared GPUs.
- Hackathon: 3 ideas recognized (top 20, 2 mentions); the looped drone became a shipped feature.
- An MCP server so AI agents can read and write the team task board (Node.js).
- Earlier (Jul–Oct 2025): a small LLM service that wrote tour narration (OpenAI API, AWS, S3).

Giraffe360 makes cameras and software for real-estate virtual tours.

EDUCATION

Riga Technical University — Professional Bachelor, Financial Engineering — 2024 – 2028 (expected)

Qualification: senior data analysis specialist. Maths, statistics and programming applied to finance.

Coursework: mathematical analysis and algebra, probability and statistics, econometrics, time-series analysis, stochastic analysis, Monte Carlo methods, portfolio analysis, actuarial mathematics, data structures and algorithms, databases/SQL, computer networks, programming (Java, C++).

AI Talent Program (Ponder) — Nov 2024 – Jan 2025

Mākslīgā intelekta talantu programma

Selective program: C++, Python, maths and AI development with industry mentors. Led to the Giraffe360 role.

Riga State Classical Gymnasium — Programming and Mathematics track

Rīgas Valsts klasiskā ģimnāzija

Specialised track in programming and mathematics (today named "Programming + AI"). Latvian state mathematics olympiads; completed the Small Mathematics University (Mazā matemātikas universitāte) at the University of Latvia.

PROJECTS

load-balancing-experiments (2026, Python) — seven reproducible experiments for an RTU networks project: DNS round robin and anycast measured from Riga, a queueing simulation of load-balancing policies ("power of two choices"), consistent / rendezvous / Maglev hashing, a mini load balancer with health checks, packet capture.

flow — offline workout tracker (2026, Dart/Flutter, private) — log sets with live timers; analysis of volume, load (ACWR), records and body measurements; custom formulas, JSON/CSV/PDF export, CI with golden-image tests. In daily use.

monte-carlo-finance (2026, Python) — Monte Carlo option pricing with variance reduction (up to 1,900x), GARCH volatility, VaR backtests with the Kupiec test, efficient frontier from 30,000 portfolios and CAPM; a study project from RTU courses.

Portfolio journey (this site) (2025–26, three.js) — a tap-driven WebGL world: RK4-integrated strange attractors, a single-stroke vector font, bloom, one continuous comet trail. No build step.

Offline Graphing Calculator (2023, with Iļja Volinskis) — graphing calculator that works offline; 4th place of 27 entries in the Riga Business School (RTU) programming competition for secondary school students (10 Jun 2023).

crypto-quant-lab (2026, Python) — 59 backtested versions of drift and volatility-targeting rules on 6 coins with a 1% fee. The source was recovered from Python 3.12 bytecode, and all 153 functions were verified.

projects_with_NEAT (2025, Python) — neuroevolution (NEAT) agents that learn Flappy Bird and Pong; a documented attempt at multi-snake survival.

RTU_program_parser (2025, Python/Selenium/pandas, with Aleksejs Peņkovs) — scrapes the RTU timetable and finds extra lectures that fit a student's own schedule; exports to Excel.

Snake + maze generator (2024–25, Python/Pygame) — DFS maze generation, dynamic terrain (lava, forest, rivers), BFS path-finding.

bezier-curve-js (2023, JavaScript) — interactive Bezier-curve drawing tool.

sort_visualization (2022, C++/SFML) — 16 sorting algorithms animated as columns, points, circles, spirals and pyramids, with shuffle patterns and speed control.

SKILLS

3D & vision	Gaussian splatting, 3D reconstruction, computer vision, depth estimation, anti-aliasing, mesh generation, diffusion-based artifact fixing, PyTorch, CUDA kernels
Research	Paper-to-production, benchmarking and evaluation
Languages	Python, C++, Dart/Flutter, JavaScript, Java, SQL
Tools	Linux, Docker, Git/GitLab CI, ClearML, AWS S3, Pygame, Selenium, pandas
Other	OpenAI API / LLM prompting, neuroevolution (NEAT), three.js/WebGL
Spoken	Latvian, English, Russian